Stream DP

Quick Guide

Version 1.0

English

1 Important Information about your Instrument

Read and follow the User Manual before using the product.

It can be accessed on the accompanying data storage device and is available for download at the following Internet address:

https://myworld.leica-geosystems.com > myDownloads

☞ Keep for future reference!

Intended use

The intended use is the data acquisition with the Stream DP system, for the detection of underground targets, such as:

- Water pipes
- Gas pipes
- Sewers
- Optic fibre
- Electric cables
- Telephone cables
- Manholes
- Cavities
- Structures in general
- etc...

The product must not be disposed with household waste.

For the Stream DP including all accessories:

⚠ WARNING

Unauthorised opening of the product

Either of the following actions may cause you to receive an electric shock:

- Touching live components
- Using the product after incorrect attempts were made to carry out repairs.

Precautions:

- ▸ Do not open the product!
- ▸ Only authorised IDS GeoRadar Service Centres are entitled to repair these products.

Trademarks

- Windows® is a registered trademark of Microsoft Corporation in the United States and other countries

All other trademarks are the property of their respective owners.

1.1 Conformity to National Regulations

1.1.1 Stream DP

EU

Hereby, IDS GeoRadar s.r.l. declares that the radio equipment type Stream DP is in compliance with Directive 2014/53/EU and other applicable European Directives.

⚠ CAUTION

This equipment is not intended for use in residential environments and may not provide adequate protection to radio reception in such environments.

UKCA

Hereby, IDS GeoRadar s.r.l. declares that the radio equipment type Stream DP is following the provisions of the applicable relevant statutory requirement S.I. 2017 No. 1206 Radio Equipment Regulations 2017.

Licensing requirements in UK can be found here:
https://www.ofcom.org.uk/manage-your-licence/radiocommunication-licences/licensed-short-range

USA

FCC ID: UFW-AE32HV
FCC Part 15

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, it may cause harmful interference to radio communications.

Operation of this equipment in a residential area is likely to cause harmful interference in which case the user is required to correct the interference at his own expense.

In order to comply with FCC RF Exposure requirements, this device must be installed to provide at least 20 cm separation from the human body at all times.

Canada

CAN ICES-003 A/NMB-003 A
IC: 8991A-AE32HV

Canada Compliance Statement
This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions:

1. This device may not cause interference
2. This device must accept any interference, including interference that may cause undesired operation of the device

Canada Déclaration de Conformité
L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes:

1. L'appareil ne doit pas produire de brouillage
2. L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement

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In order to comply with FCC / ISED RF Exposure requirements, this device must be installed to provide at least 20 cm separation from the human body at all times.

Others

The conformity for countries with other national regulations has to be approved prior to use and operation.

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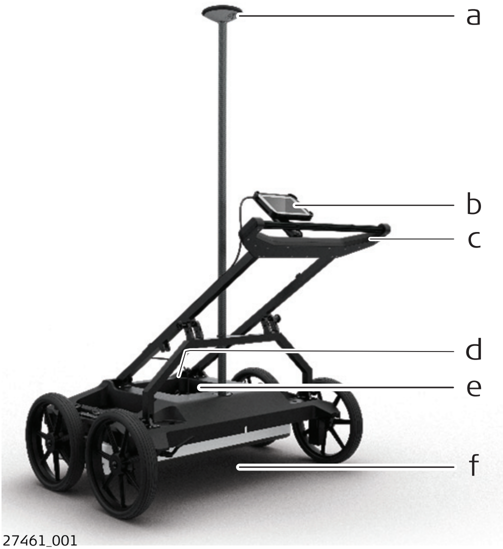
2 Description of the System

General

Stream DP is a compact GPR massive array system. It is composed of a single antenna fixed to a compact and manoeuvrable frame. Stream DP is used for collection of GPR survey over large areas, mainly for underground utility mapping and pipe detection. Also, any type of buried objects or structures may be the aim of the system, if these are buried within its depth of reach.

2.1 System Components

Overview



a GNSS antenna pole support
GNSS is not included.

b Laptop holder
PC Laptop is not included.

c Handle and rudder

d Stream DP main body
Including frame, wheel encoders (two), wheels (four big wheels and two pivoting wheels)

e Antenna top with battery compartment
Battery charger comes in a separate box.

f Antenna bottom with replaceble sledge

Sub-components

The system can be easily disassembled in these sub-components:

27462_001

a	Handle and rudder
b	Main body (frame)
c	Antenna
d	Four wheels and two pivoting wheels

3 Technical Data

Environmental specifications

Temperature

Type	Operating temperature	Storage temperature
Stream DP	−20 °C to +50 °C (−4 °F to +122 °F)	−40 °C to +60 °C (−40 °F to +140 °F)

Protection against water, dust and sand

Type	IP class (IEC60529)	Pollution degree (IEC61010-1)
Stream DP[1)]	IP65	2

Humidity

Type	Protection
All instruments	Max 95% non-condensing. The effects of condensation are to be effectively counteracted by periodically drying out the instrument.

Weights

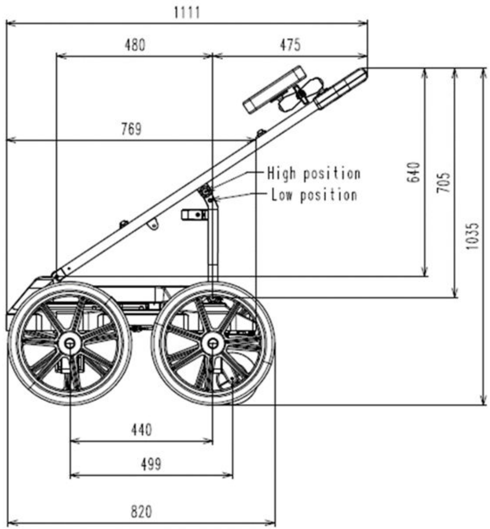
Product	[kg]
Stream DP	42.0
Each part of the system	< 20.0

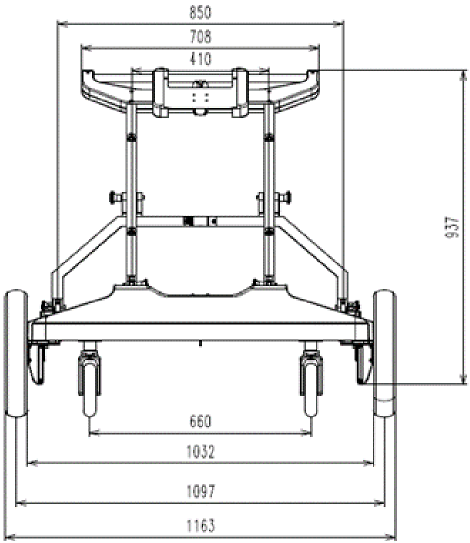
[1)] Valid only during operation

Dimensions

Type	[mm]
Stream DP – full	1163 × 820 × 1111
Antenna	1022 × 521 × 219

Side view	Rear view





4 Care and Transport

Transport in the field

When transporting the equipment in the field, always make sure to carry the product in its original container.

Transport in a road vehicle

Never carry the product loose in a road vehicle, as it can be affected by shock and vibration. Always carry the product in its container and secure it.

For products for which no container is available use the original packaging or its equivalent.

Shipping

When transporting the product by rail, air or sea, always use the complete original IDS GeoRadar packaging, container and cardboard box, or its equivalent, to protect against shock and vibration.

Shipping, transport of batteries

When transporting or shipping batteries, the person responsible for the product must ensure that the applicable national and international rules and regulations are observed. Before transportation or shipping, contact your local passenger or freight transport company.

5 Operation

5.1 Instrument Setup

Overview

Refer to the Stream DP Installation Manual for the listed topics:

- General description of the hardware
  - Mechanical structure
  - Lifiting system
  - Radar system, antenna modules
  - Connection and ports
  - Battery
  - Wheel types
  - Odometer, wheel encoder
  - Wheel blockage system
  - Adjustable and removable handle
  - GNSS support pole
  - Controller specifications
  - Laptop support
  - How the system is delivered and packed
- How to assemble the system and safety guidlines
- How to set up the Stream DP in transport mode
  - Folding the handle
  - Disassembling the system
- Spatial positioning in Stream DP
  - How to connect a Smart GPS
  - How to connect a GNSS
  - How to connect a TPS

5.2 Assembly

Step-by-step

1. Fix the four wheels on the main body using the wheel button.

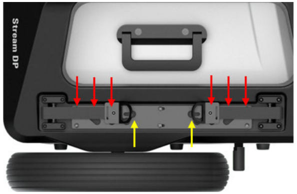
2. Mount the handle by fixing six pins:
   - Four short pins on the main frame

   - Two long pins on the central rudder

3. Place the antenna on a horizontal surface. Move the frame over the antenna by fitting the frame in the antenna's inserts.

4. Secure the antenna to the lifting system fastening the four hooks.

5. While lifting the antenna using the handle, insert each knob in one of the holes.



At the same time, adjust the height from the soil using the lifting system.

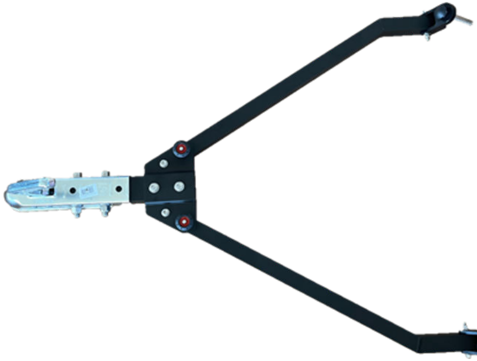
6. Install the tablet on the support.
7. Install the GNSS pole support.
8. Connect the LAN cable from the antenna to the laptop port.
9. Connect the two-wheel encoder cables to the two-"wheel" connectors and turn 1/4 to fasten.

Stream DP in operative mode

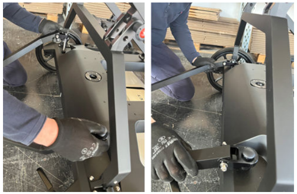
5.3 Assembly of the Towing Kit (Optional)

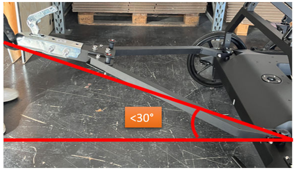
Step-by-step

The towing kit (974856) can be purchased optionally.

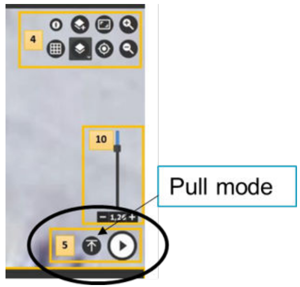


Operator outside the vehicle:

1.	Remove the two central pins on both arms of the towing kit.	
2.	Insert the two long pins in the frame holes and fix the free arms to the Stream DP frame.	

3.	Insert the two central pins again to block the aperture of the towing kit.	
4.	Connect the towing kit to the towball (ISO 50 standard) of the vehicle. ☞ The angle between Stream DP and the vehicle must not exceed 30°.	
Operator inside vehicle:		
5.	Option: Plug the long LAN cable to the laptop holder.	

6. In the uMap acquisition software, select the Pull mode.



WARNING

Moving and/or towing of the product

IDS GeoRadar makes no guarantee that the moving and/or towing of the product/vehicle complies with the regulations in force in the country/area of use.

Precautions:

- Adhere to or comply with, all national laws, regulations and conditions for the operation of the product/vehicle in force in the country and/or the site owners.
- The user is responsible for the evaluation of the product/vehicle installation, location and the adaptation to the applicable vehicle. It is the users responsibility to decide on an installation plan to reduce potential hazards. IDS GeoRadar is not liable for any damages caused by incorrect installation and/or product misuse.
- Follow all regulations regarding warnings, signalling, and lighting of the product/vehicle and any other suitable safety measures.

5.4 Power On the System

Step-by-step

1.	Open the battery compartment by loosening the screw. Then insert up to two batteries in the compartment.	
2.	Turn the system on.	
3.	Launch uMap software on the data logger.	

6 Appendix A

Disclaimer

1. General

a) The present Disclaimer applies to all products designed, produced and distributed by IDS GeoRadar S.r.l., its Subsidiaries, Affiliated and authorized Distributors (the "Products"). IDS GeoRadar S.r.l. reserves full ownership and intellectual property rights of any "Information" contained in this Disclaimer including Trade Marks and Graphics. No part of this Disclaimer may be used or reproduced in any forms without the prior written agreement of IDS GeoRadar S.r.l.

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2. Initial Precautions for Setting-up and Use of the Products

a) The User/Buyer, for setting-up and using the Products, shall consult the official documentation provided by IDS GeoRadar S.r.l. for the Products ("Reference Documentation") and carefully ascertain the compliance with national laws and requirements, which may limit or even forbid their use.

b) For Products which shall operate by circulation in Public Areas/Roads, with or without moving traffic, Buyer/User shall verify the approval of local authority and/or site's owner according to their specific procedures. IDS GeoRadar S.r.l. shall not be liable for any direct, indirect, special, incidental or consequential damages or injuries, including without limitation, lost revenues or lost profits, resulting by un-authorized use of the Products in Public Areas/Roads.

c) Buyer/User further warrants:
   - That these Products are not being used, in the design, development, production or use of chemical, biological, nuclear ballistic weapons. Buyer/ User will defend, indemnify and hold IDS GeoRadar S.r.l. harmless against any liability (including attorney's fees) for non-compliance with the terms of this article.
   - That, if IDS GeoRadar S.r.l. requires that Buyer/User shall carry out a training with reference to some Product categories, no operation or use of the Products shall be started before its designated Operator/s has got the User Certificate, as defined by IDS GeoRadar S.r.l. specific procedure which the Buyer confirms to know and accept.

d) For Products which include specific "Operational" software with automatic data processing and analysis "Tools", e.g. the **IBIS Products and Hydra Products**, User shall be aware that the results provided by these "Tools" **may be not error free**. User that completely relies on the outcomes provided by these Tools only, does it at his own risk.

e) In no event IDS GeoRadar S.r.l. shall be liable for special, direct, indirect, incidental, exemplary, punitive or consequential damages including, but not limited to, loss of profits or revenue, caused by the use of the Products, either separately or in combination with other products or relied upon the results provided by the above "Tools".

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a) The User shall follow the instructions provided by IDS GeoRadar S.r.l. in its official "Reference Documentation" for the Product, in particular the User's Technical Manual which contains all the specific steps and recommendations for a correct setting-up and use of the Product.

b) In no event IDS GeoRadar S.r.l. shall be liable for special, direct, indirect, incidental, exemplary, punitive or consequential damages including, but not limited to, loss of profits or revenue, caused by the missed or incomplete observance of the instructions and prescriptions for the use of the Products, either separately or in combination with other products, including but not limited to the following main aspects:
    a) Use of IDS GeoRadar S.r.l. Products outside its limitation of use, without proper and adequate scientific/technical knowledge or without specific training.
    b) Use of results/outcomes of the measurements performed by the Product aimed to safety aspects without using adequate control procedures and assessment by skilled personnel.
    c) Opening of the Equipment (for HW Products) without express written authorization of IDS GeoRadar S.r.l.;
    d) Unauthorized changes and additions to the Products.
    e) Use of the Products connected to suspected non-working equipment or with equipment (mainly PC) having characteristics not in compliance with the required specifications of IDS GeoRadar S.r.l. not expressly authorized by IDS GeoRadar S.r.l.;
    f) Poor or faulty operation of the electrical and telecommunication networks not directly managed by IDS GeoRadar S.r.l. or its delegates..
    g) Poor or faulty operation Software/Hardware of the third parties connected with IDS GeoRadar S.r.l. Equipment.
    h) Poor or faulty operation of the Products due to Software Virus which infected the Products after their delivery.

i) Use of the Products which have encountered suspected manumissions, accidents, electrostatic shocks, flashes, fire, earthquake, flooding or other natural disasters or unexpected events.
j) Use or storage of the Products outside the limits of the “Operational Temperature Range” specified by IDS GeoRadar S.r.l.

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Leica Geosystems AG
Heinrich-Wild-Strasse
9435 Heerbrugg
Switzerland

www.leica-geosystems.com